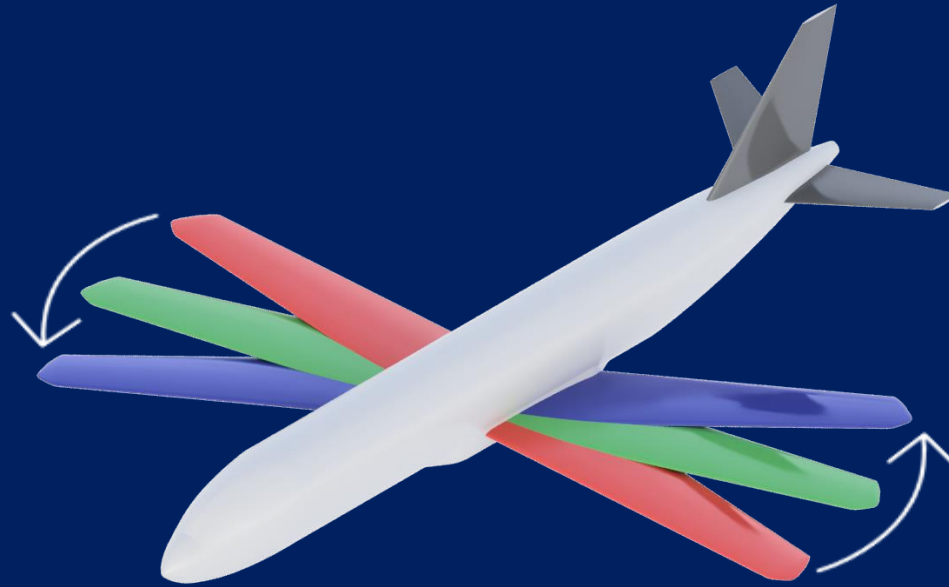




TAMING THE OBLIQUE WING:

Improving Fuel Efficiency by Developing and Flight Testing an Oblique Wing Aircraft Utilizing a Novel Control Method

Kevin Shen | Olympia, WA

- The oblique wing is a type of variable-sweep wing positioned at a slanted (oblique) angle across the aircraft [4].
- **The oblique wing generates lower drag than conventional aircraft** [4], resulting in reduced fuel costs [3] and lower environmental impact [5].
- **Reduced wave drag** [6] at transonic speed by distributing lift over a longer distance at high sweep angle, as seen in Figure 2.

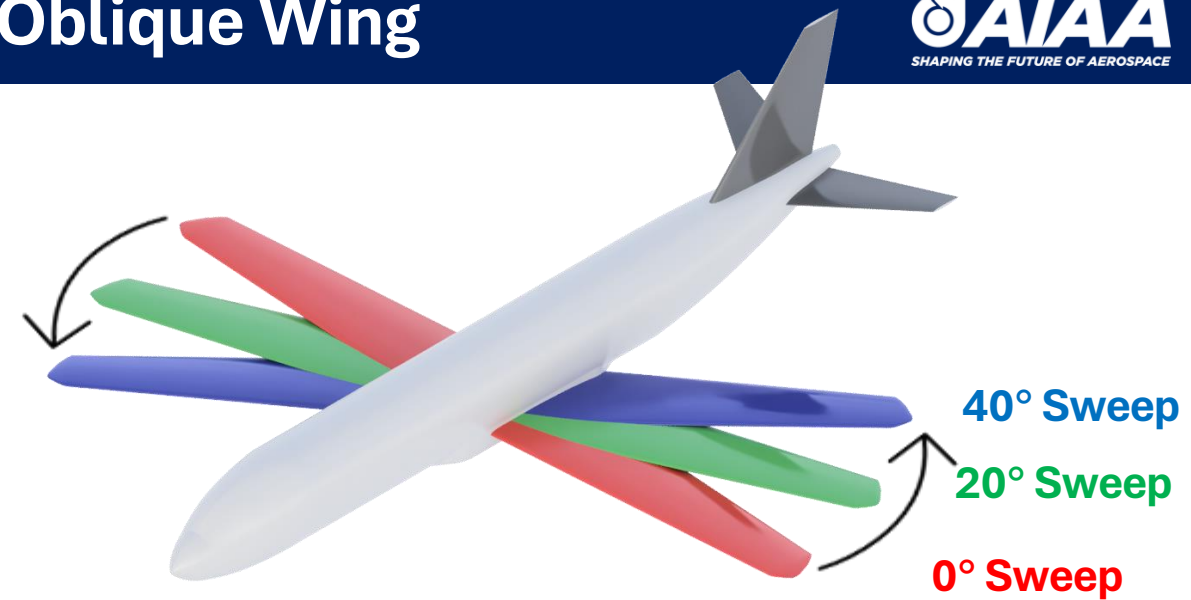


Figure 1. The variable-sweep oblique wing at various sweep angles. *Image created by Kevin Shen.*

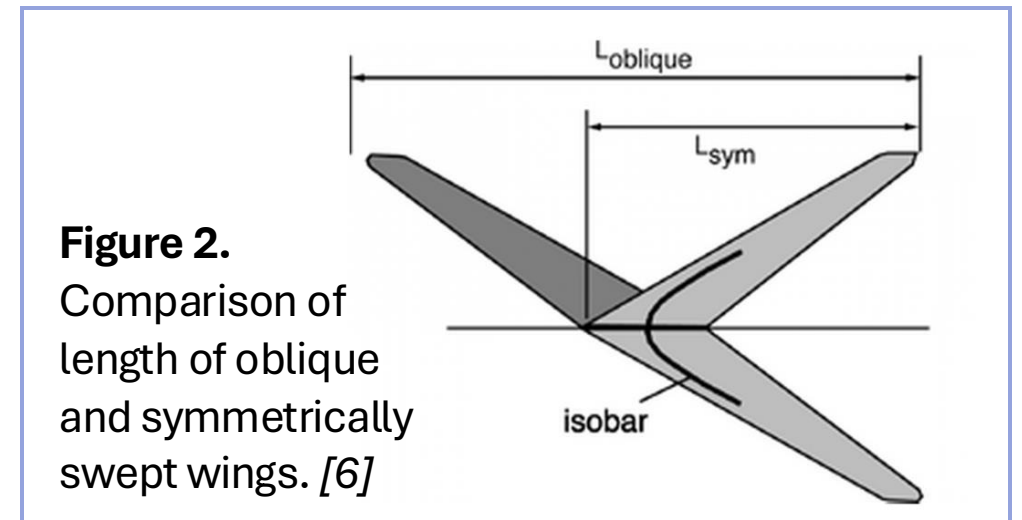
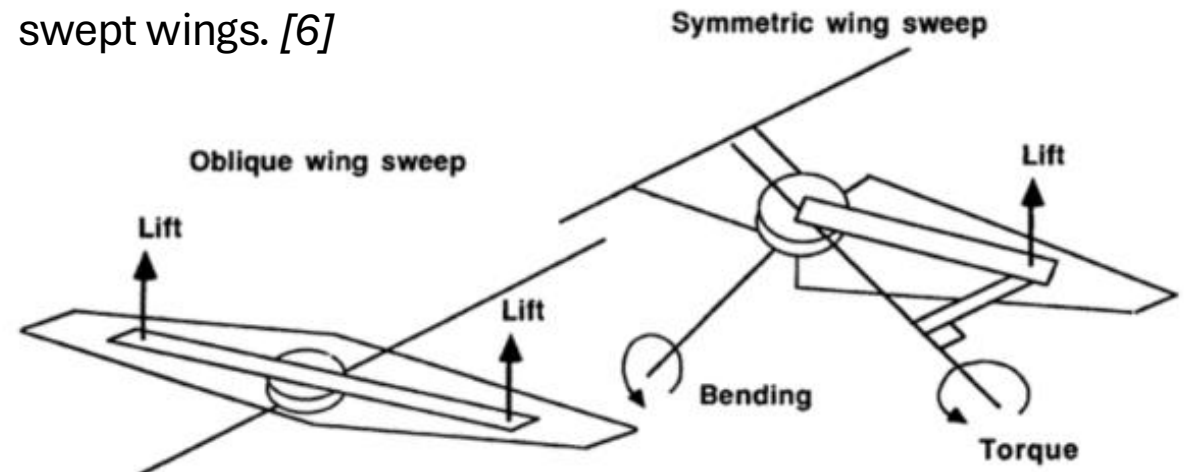


Figure 2. Comparison of length of oblique and symmetrically swept wings. [6]

- **Reduced induced drag** [6] at subsonic speed due to higher aspect ratio when at low sweep angle.
- **Structurally simpler** and thus lighter than conventional variable sweep wings because oblique wings only require one pivot, do not take bending or torque loads, and can be built with a single straight wing spar [4].

Figure 3. Structural advantage of oblique over symmetrically swept wings. [6]



- Aerodynamic coupling is caused by the oblique wing's **asymmetry**.
- Each side of the wing reacts differently to changes in roll, pitch, or yaw, “linking” or “coupling” axes together, making the aircraft **difficult to fly**.
- **Must be significantly reduced** before oblique wings can be implemented.

Figure 4. Aircraft axis and controls on an oblique wing. *Image created by Kevin Shen.*

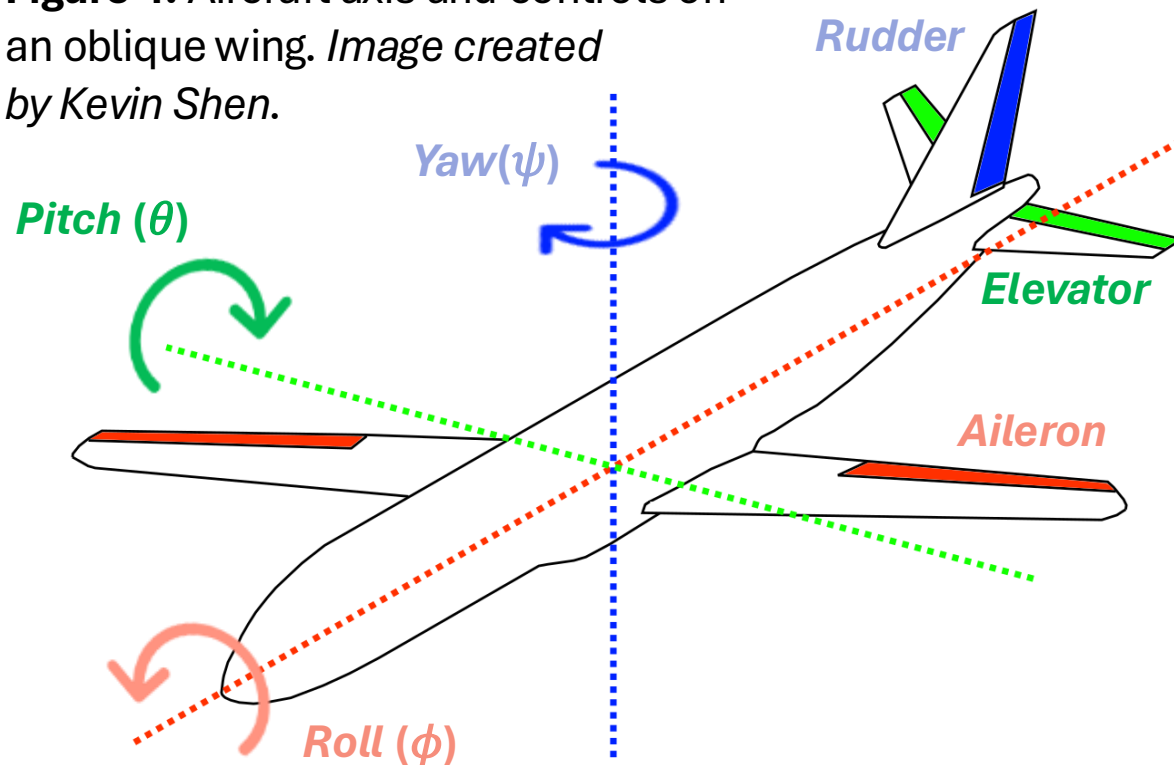
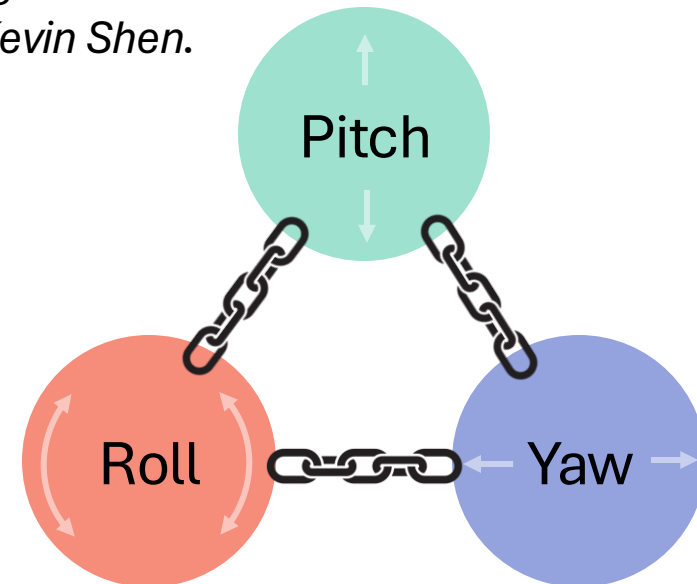


Figure 5. Illustration of aerodynamic coupling, where motion in pitch, roll, and yaw are “chained” or “coupled” together. *Image created by Kevin Shen.*



- In the 1970s, NASA developed the AD-1 research aircraft, the only piloted oblique wing ever built. However, it lacked flight computers and pilots reported it was difficult to fly due to aerodynamic coupling.
- The AD-1 program was shut down in 1982 due to budget cuts, **leaving control challenges unresolved.**



Figure 6. NASA's AD-1 research aircraft. [4]

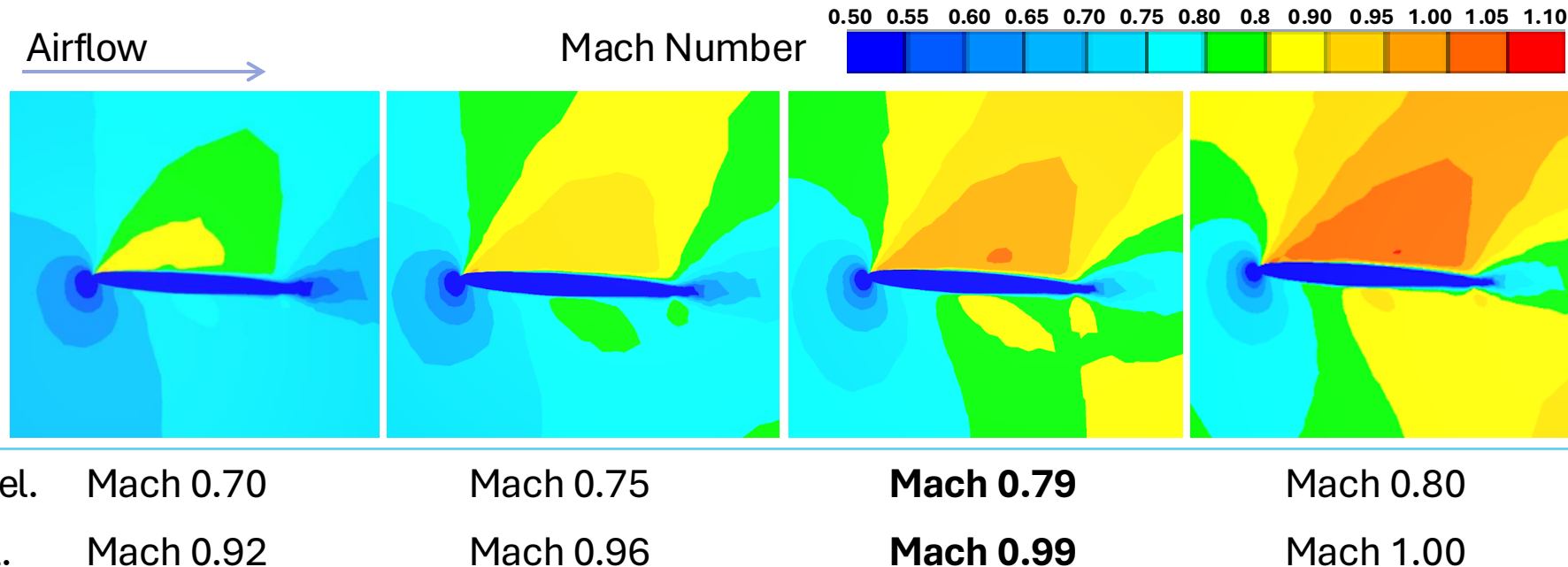
Problem: Oblique wing aircraft are more fuel-efficient than conventional aircraft but are **challenging to fly** due to **aerodynamic coupling**, preventing implementation.

Solve the problem by:

1. Designing a modern (transonic) **oblique wing aircraft** and analyzing its efficiency in comparison to a conventional, symmetric design to quantify reductions in emissions and fuel expenditures.
2. Leveraging modern computers to develop a novel **anti-coupling flight computer** and reduce the previously unsolved aerodynamic coupling issues of oblique wings, improving safety and control.
3. Combining the aerodynamic design and flight computer into a **flyable test model** to physically test aerodynamic coupling and control characteristics.

1 2D Computational Fluid Dynamics (CFD) of the SC(2)-0610 airfoil [2] at 4° angle of attack at various airspeeds to find critical Mach number M_{crit} .

Figure 7.
Airfoil CFD simulations.
Image created by Kevin Shen.



2 Preliminary sweep angles found quickly using sweep theory [1].

$$M_{crit} = M_{\infty} \cos(\lambda)$$

M_{crit} = critical Mach number of airfoil,
 M_{∞} = freestream airspeed in Mach,
 λ = optimal wing sweep angle

Equation 1. Optimal sweep formula.

- 3 Sweep angles refined using slower, but more accurate 3D CFD analysis. Determined that the entire aircraft's M_{crit} at 30° sweep was **Mach 0.82**.

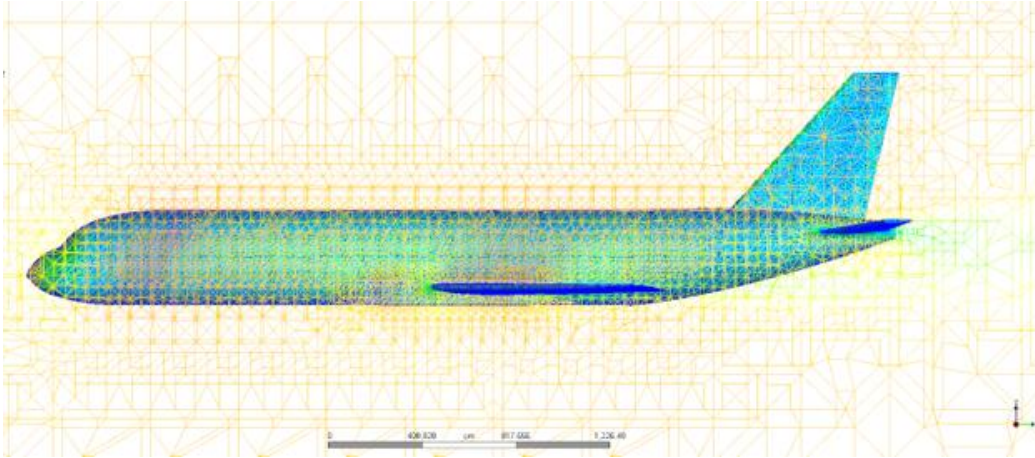


Figure 8.CFD Mesh (approx. 5M elements). *Image created by Kevin Shen.*

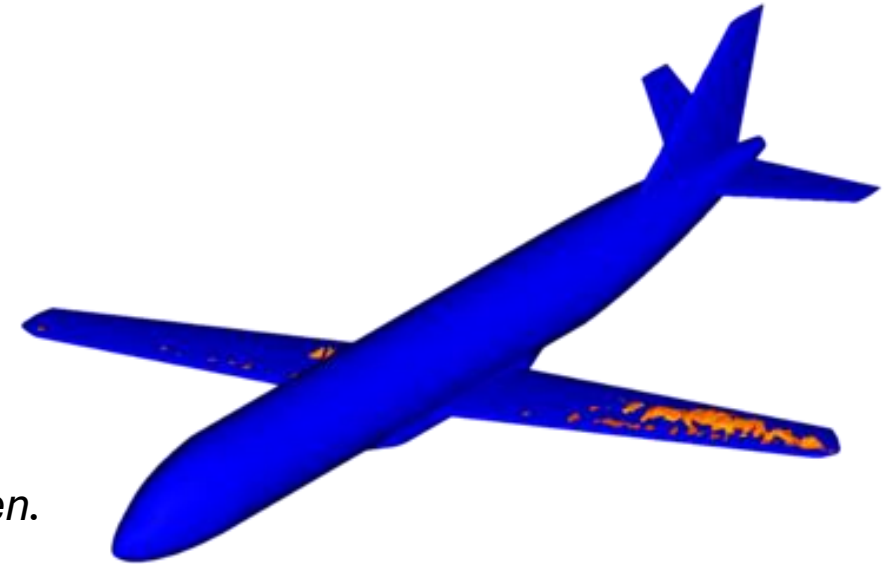


Figure 10. Identifying locations of supersonic flow over the oblique wing and adjusting in the next iteration. *Image created by Kevin Shen.*

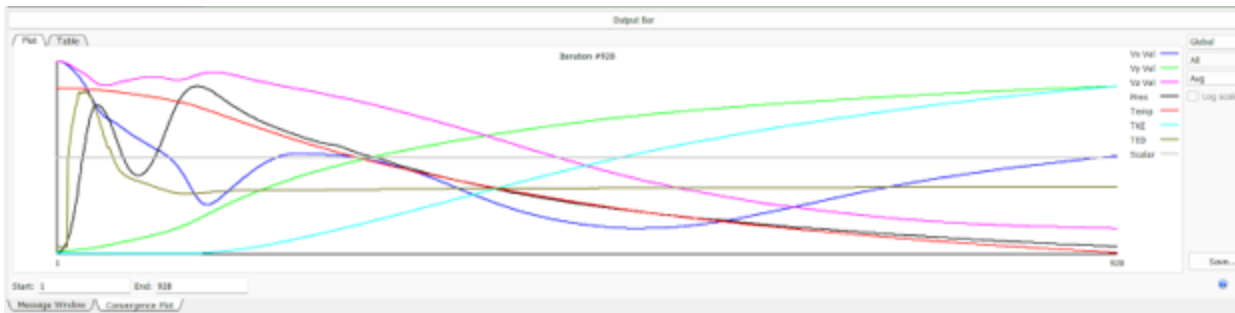
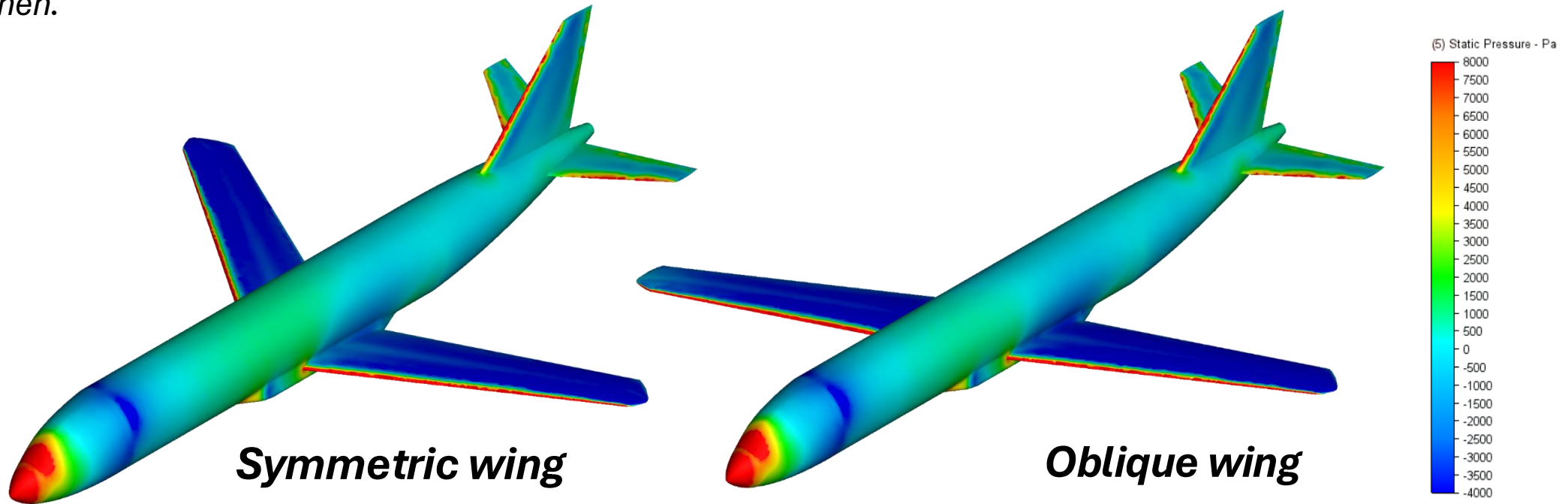


Figure 9. Full 3D CFD Convergence Plot. *Image created by Kevin Shen.*

- Purpose: Use CFD to quantify the potential drag reduction of a full-scale oblique wing aircraft to validate the design's fuel efficiency benefits and justify the additional complexities of an oblique wing aircraft.

Figure 11. Comparing the conventional, symmetric wing design (left) against the oblique wing design (right). Conditions set to Mach 0.82 at 10,000m altitude. Sizing based on the A320 aircraft. *Image created by Kevin Shen.*



- **9.2% reduction in drag** between the oblique wing and the symmetric wing at 30° sweep under cruise conditions, given both wings produce the same lift force.
- Aligned with previous literature that predicted an 8.95% reduction in drag between an oblique wing and a symmetric wing at 30° sweep [6] (though at a slightly different airspeed).
- A 9.2% reduction in drag results in an approximately 9.2% reduction in fuel burn, with significant implications (shown below).

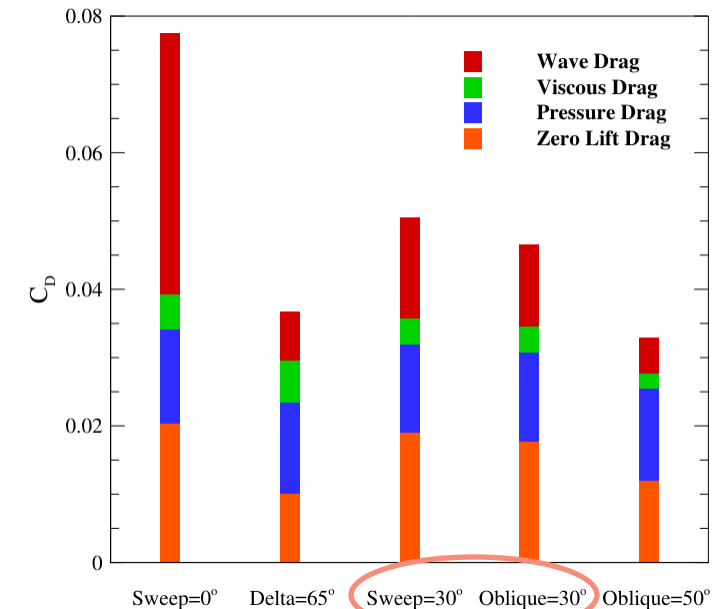


Figure 12. Previous estimates of efficiency improvements. [6]

Potential Annual Benefits of Oblique Wing Implementation



96 million tonnes of CO₂ emissions globally

(Estimated by multiplying % fuel reduction with current total aviation emissions [5]).



25 billion dollars in total aviation fuel costs

(Estimated by multiplying % fuel reduction and current total aviation fuel costs [3]).

Parameter	Value	Parameter	Value
Thrust-to-Weight Ratio at Full Thrust	1.5:1	Maximum Oblique Wing Sweep	30°
Weight-to-Surface Ratio	5 kg/m ²	Wingspan	1.1 m
Maximum Takeoff Mass	0.48 kg	Tip-to-tail Length	1.07 m
Static Stability Margin (AC to CG)	12.5%	Root Chord	0.156 m
Aspect Ratio	9	Tip Chord	0.095 m
Taper Ratio	0.5	Wing Area	0.138 m ²
Horizontal Stabilizer Area	0.037 m ²	Vertical Stabilizer Area	0.027 m ²

Maximum required sweep of 30° based on CFD sweep optimization

1.1m wingspan

Wing pivot at aerodynamic center

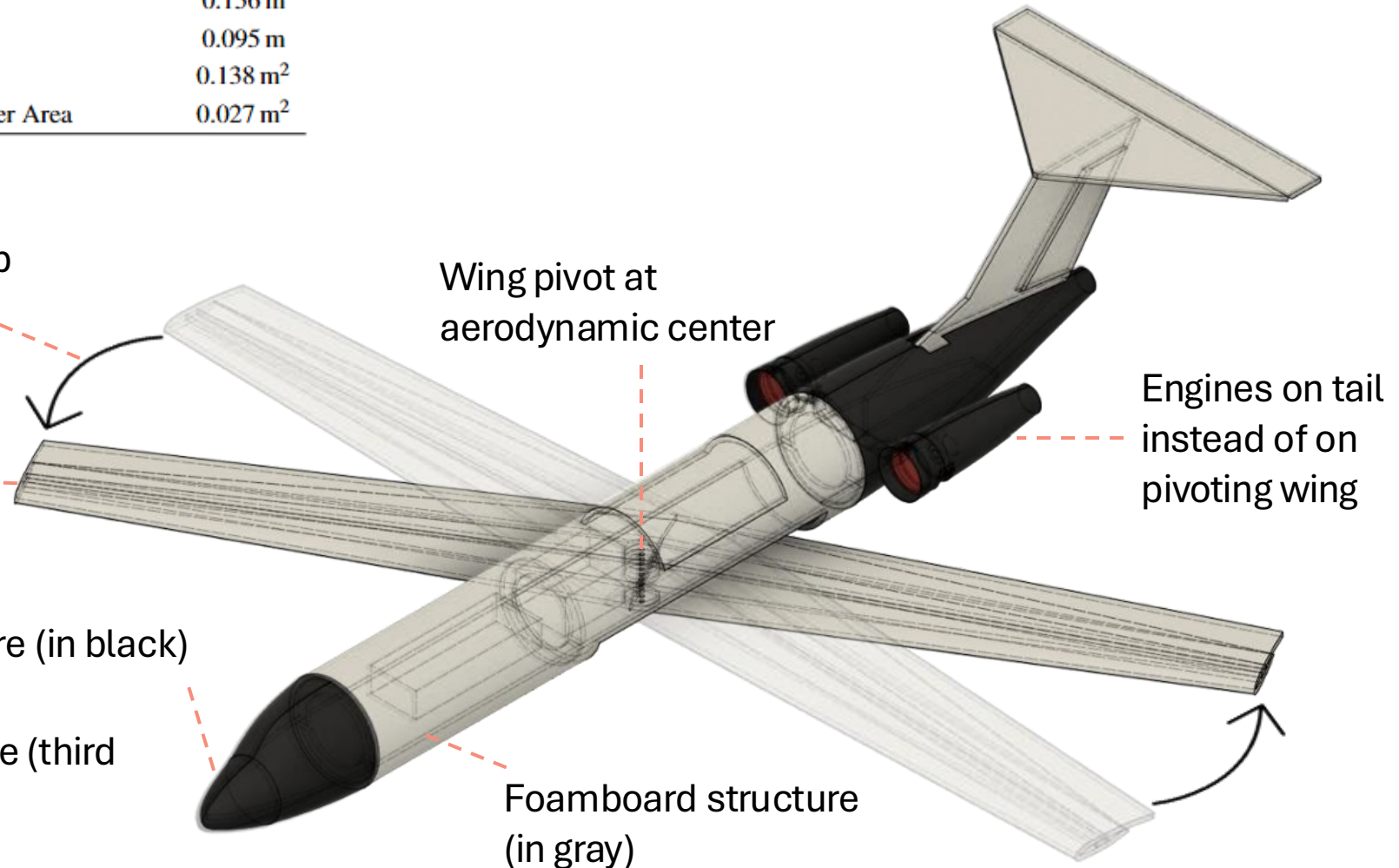
Engines on tail instead of on pivoting wing

3D printed structure (in black)

Foamboard structure (in gray)

Figure 13. The CAD model of the final airframe (third prototype).

Image created by Kevin Shen.



- Pivot actuates the wing between 0 and 30 degrees of wing sweep while supporting the lift forces of the oblique wing.
- Located at the aerodynamic center of the wing.
- 3D printed.
- Only two parts (excluding servo), so it is simple and lightweight.

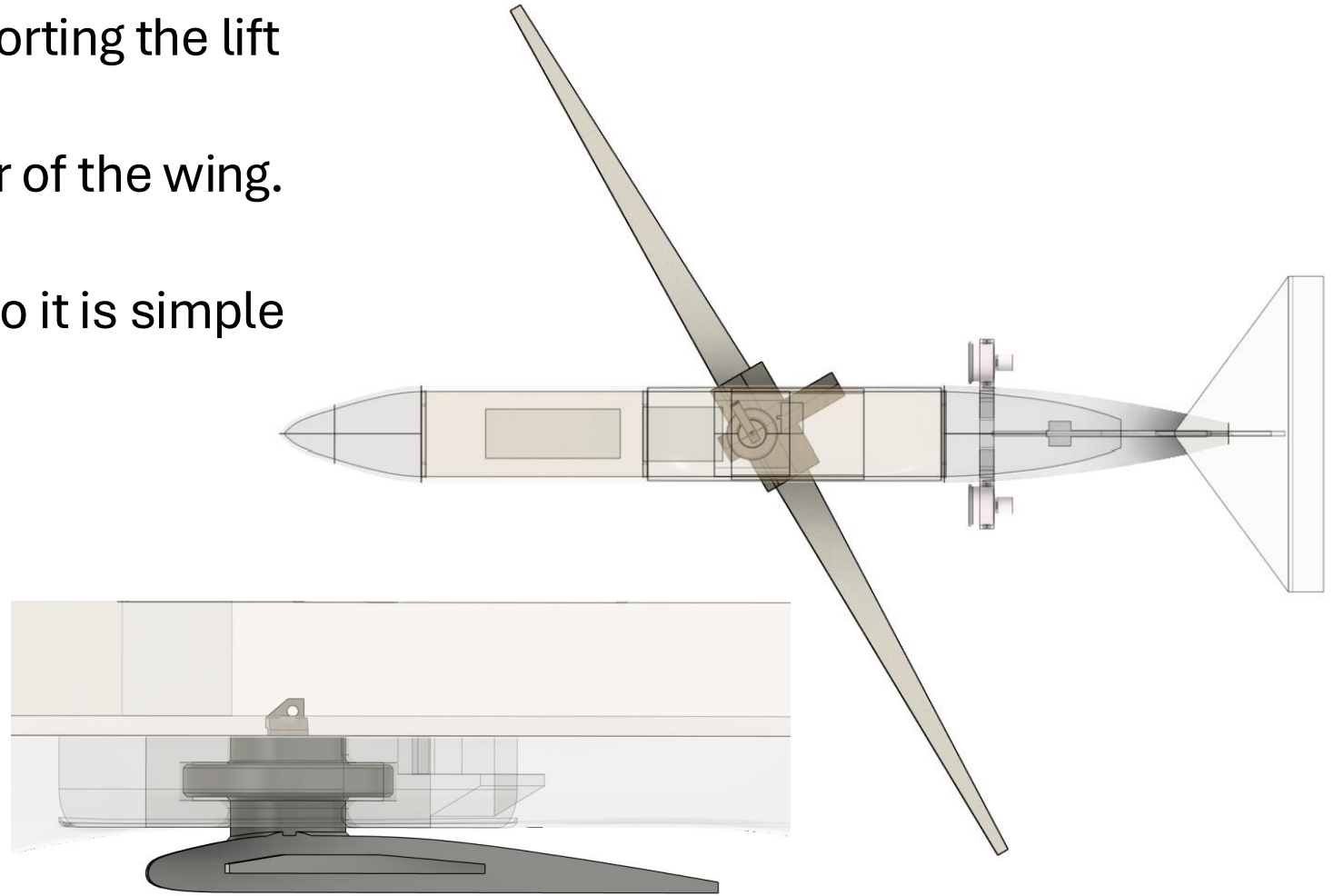
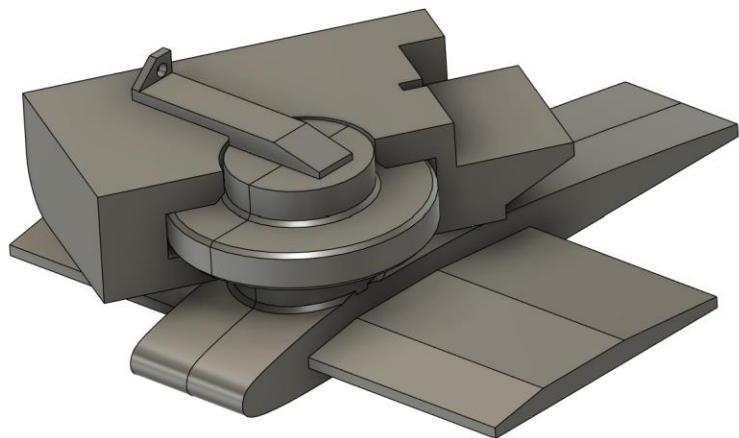


Figure 14. Model wing pivot CAD models. *Image created by Kevin Shen.*

- Custom designed PCB to connect processor, IMU, servos, radio input, etc.
- Included custom data logging system to record flight data.
- Based around a Teensy 4.1 microcontroller.
- Used dRehmFlight for low-level servo control, built on top using custom C++ software.

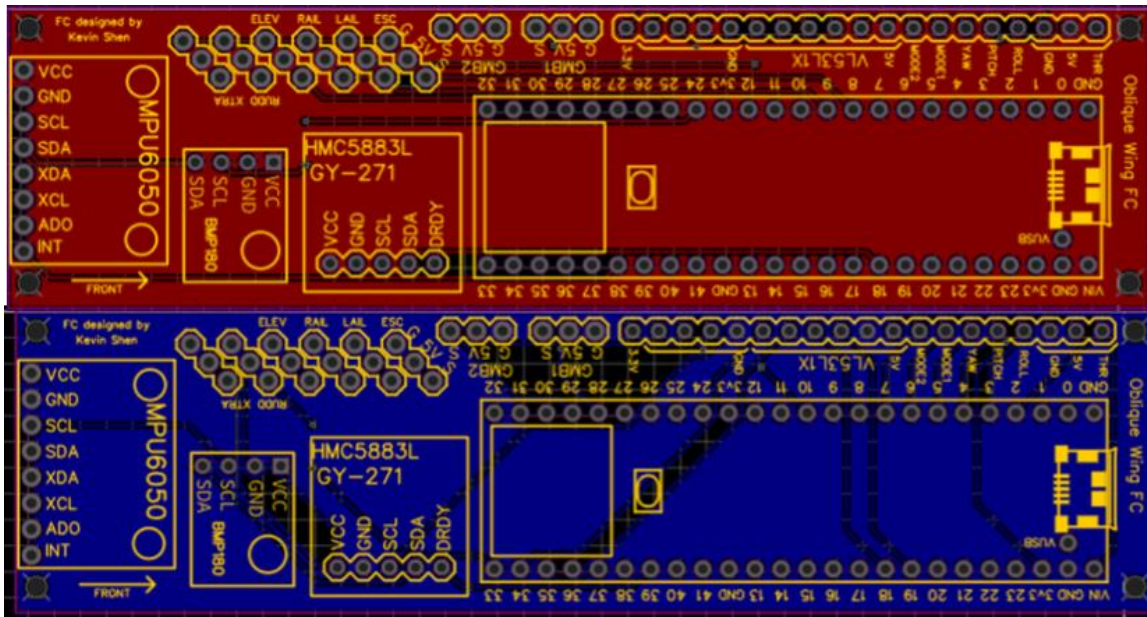


Figure 15. Top and bottom layer of the flight computer's PCB. *Image created by Kevin Shen.*

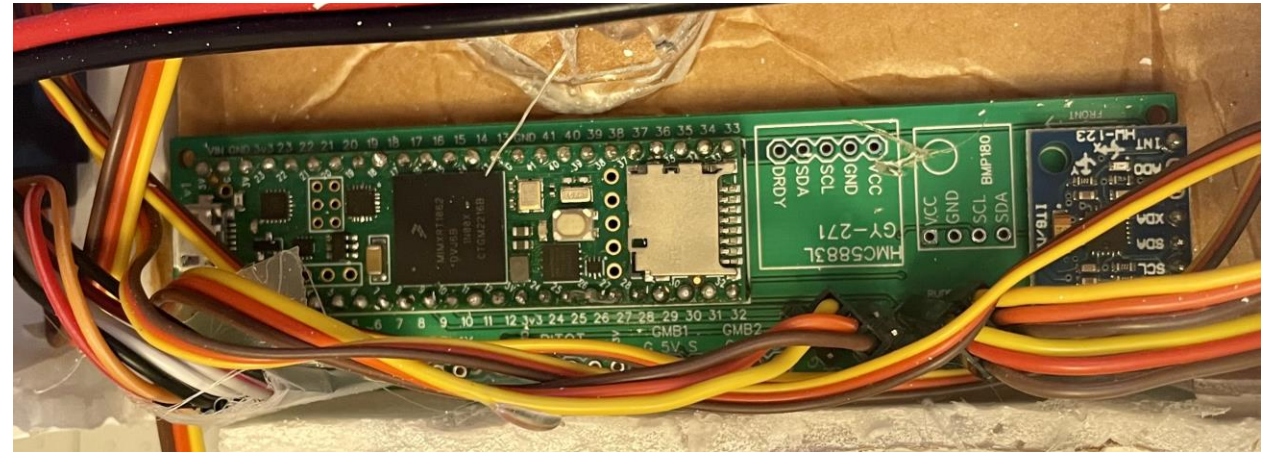


Figure 16. The custom-designed flight computer inside the airplane, soldered by hand. *Image created by Kevin Shen.*

Flight-Capable Test Vehicle: Photos



Figure 17. Ground test with 0° and 30° sweep superimposed. *Image created by Kevin Shen.*

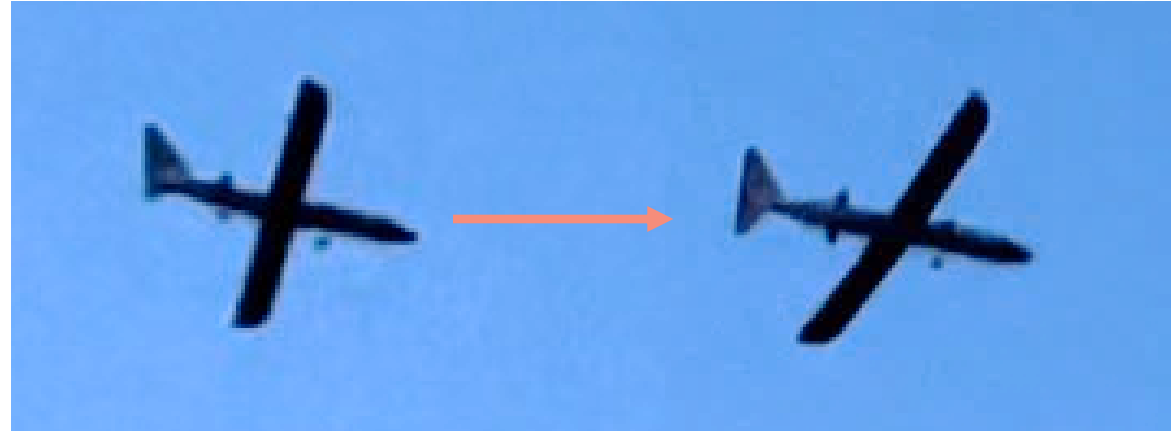


Figure 18. The flight model changing sweep in flight. *Image taken by Kevin Shen's parent.*



Figure 19. Hand launching and remotely piloting the model. *Image taken by Kevin's parent.*

- Goal: Design a function that inputs pilot roll command P_ϕ , pilot pitch command P_θ , pilot yaw command P_ψ , measured roll rate $\frac{d\phi}{dt}$, measured pitch rate $\frac{d\theta}{dt}$, and measured yaw rate $\frac{d\psi}{dt}$ and outputs aileron command C_a , elevator command C_e , and rudder command C_r , such that the pilot has full control authority over all three axes with minimal aerodynamic coupling between axes.

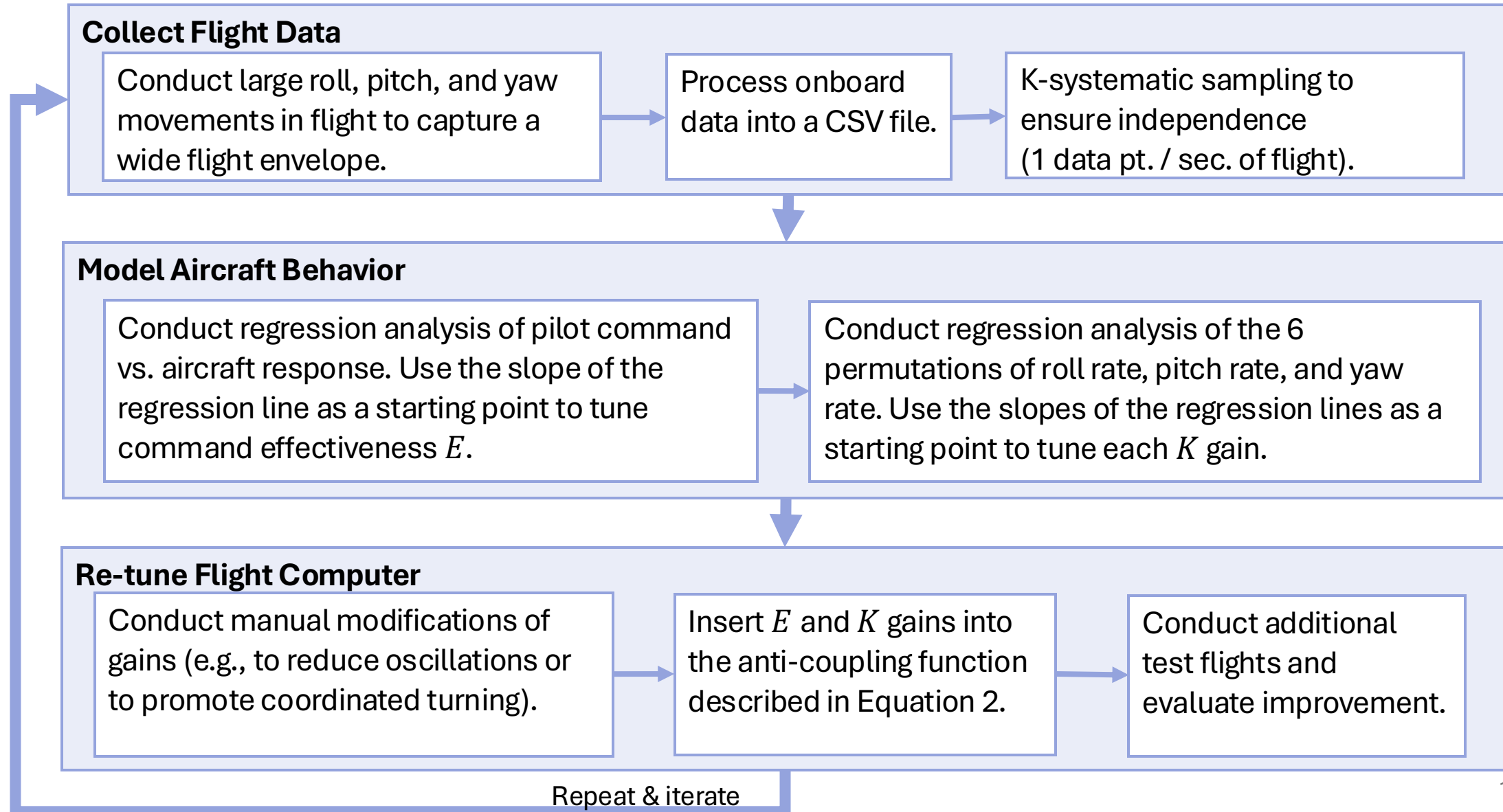
Equation 2. The final anti-coupling function.

Equation created by Kevin Shen.

$$\begin{bmatrix} C_a \\ C_e \\ C_r \end{bmatrix} = \begin{bmatrix} P_\phi \\ P_\theta \\ P_\psi \end{bmatrix} + \begin{bmatrix} E_\phi & 0 & 0 \\ 0 & E_\theta & 0 \\ 0 & 0 & E_\psi \end{bmatrix} * \begin{bmatrix} 0 & -K_{\theta\phi} & -K_{\psi\phi} \\ -K_{\phi\theta} & 0 & -K_{\psi\theta} \\ -K_{\phi\psi} & -K_{\theta\psi} & 0 \end{bmatrix} * \begin{bmatrix} \frac{d\phi}{dt} \\ \frac{d\theta}{dt} \\ \frac{d\psi}{dt} \end{bmatrix}$$

Output signal to control surfaces Add in pilot input Scale each axis based on its control effectiveness (E gains) Negate aerodynamic coupling model gains (K gains) to remove coupling Attitude rates

- Goal: Make a method of tuning the gains in the anti-coupling algorithm to effectively remove aerodynamic coupling.



- Statistically analyzing the coupling relationship between roll, pitch, and yaw.
- Two-pronged approach:
 1. Hypothesis tests (two-sample t-tests) to determine the statistical significance of coupling relationships.
 2. Regression analysis to calculate R^2 value between axes to quantify the coupling strength between axes.
- Create a relative test between flights **without** the flight computer active (control group) and flights **with** the flight computer active (experimental group)

- First, ensuring normality in the data to run the statistical test via Q-Q plots:

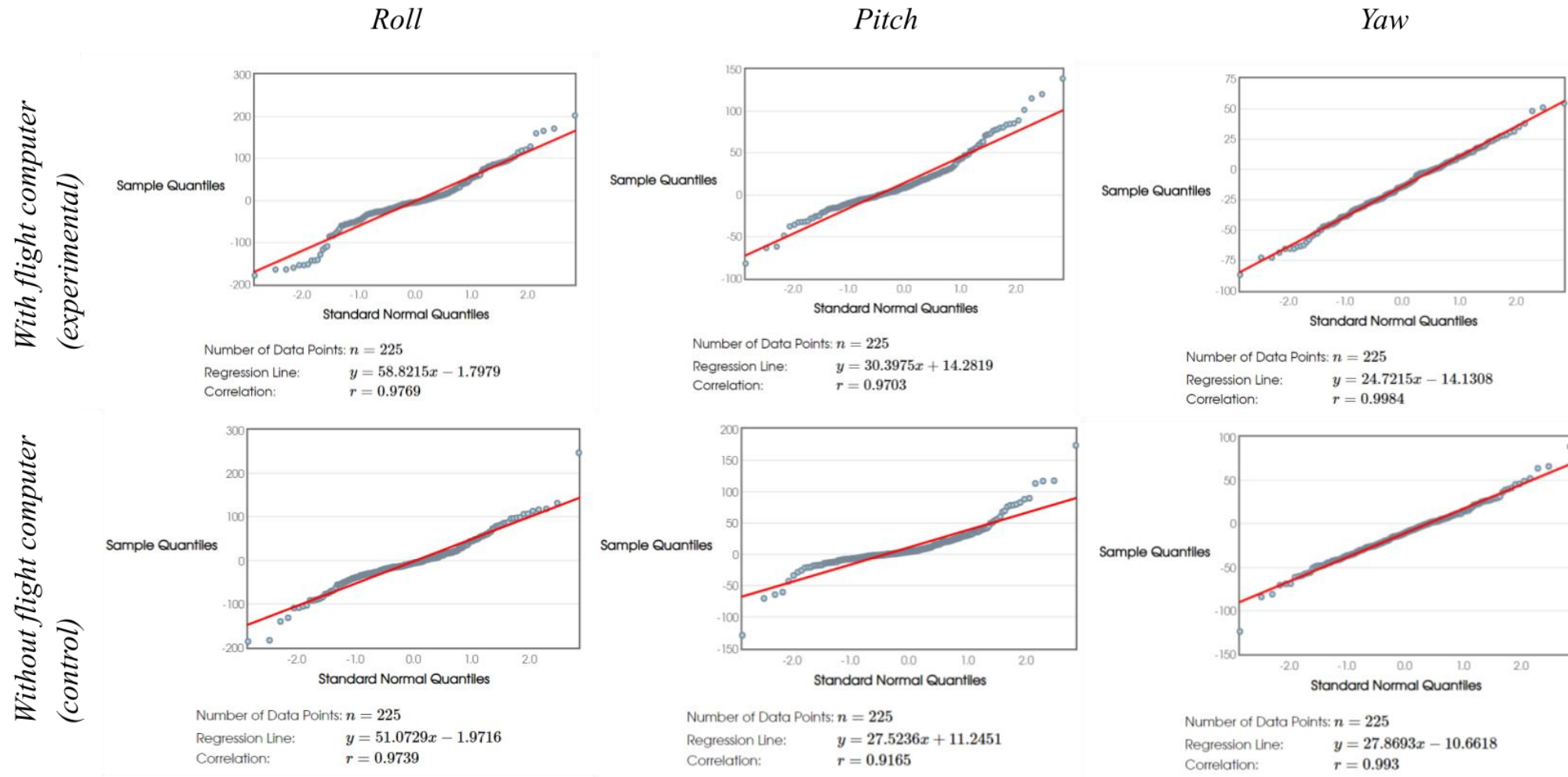
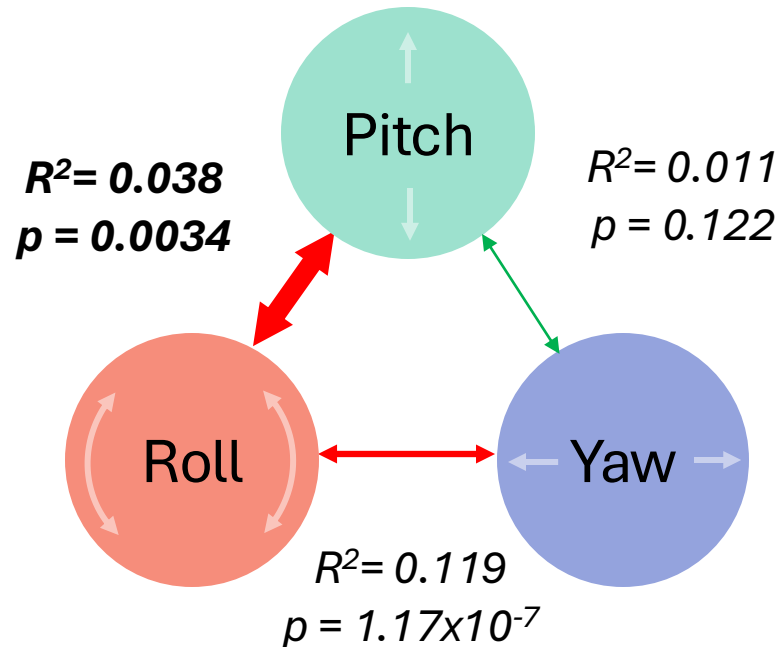


Figure 20. Q-Q plots to ensure normality. *Image created by Kevin Shen.*

Without Flight Computer



With Flight Computer

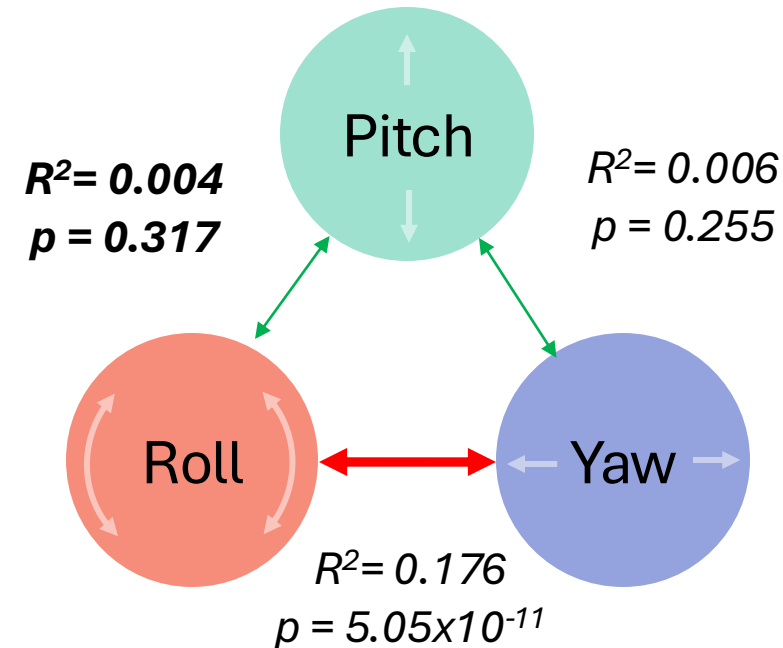


Figure 21. Graphical representation of statistical analyses of coupling behavior.
Image created by Kevin Shen.

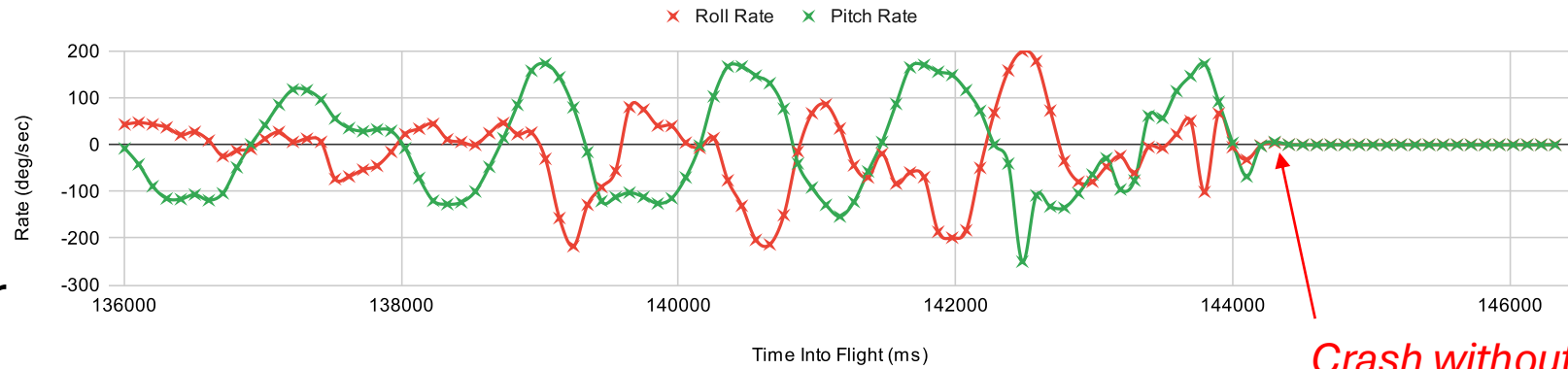
Explanation for graphical representation of aerodynamic coupling:

- A comparison is made between data collected from flights without the flight computer (left) and flights with the flight computer (right).
- The thickness of each arrow is proportional to the coupling strength as R^2 between with and without flight computer.
- The color represents hypothesis testing results (red signifies a statistically significant relationship between the two variables, as the slope of the regression line relating the two is significantly different than 0 at significance level $\alpha = 0.05$).
- Data were trimmed to $n = 225$ for both datasets to ensure a fair comparison.

- Flight data from tests without the flight computer **revealed** that the major aerodynamic coupling issue to overcome was a statistically significant, **inverse coupling relationship between roll and pitch**.
- With a flight computer and the novel anti-coupling algorithm, roll-pitch coupling was **reduced by 99%** (using R^2 as a metric) **to statistically insignificant levels**, an improvement over the previous NASA AD-1 aircraft.

- Graphing aircraft motion in the time domain to visualize and confirm the aerodynamic coupling behavior suggested by statistical analysis.

**Without
flight
computer**



*Crash without
flight
computer*

**With
flight
computer**

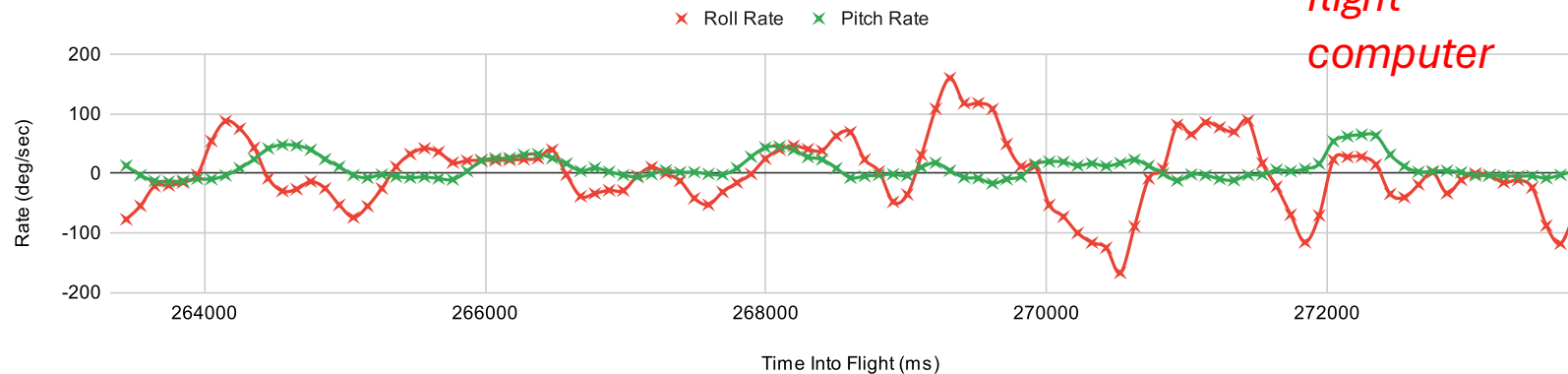


Figure 22. Excerpts of flight data: Roll and pitch versus time. Without the flight computer activated (top), there is a strongly coupled, inverse roll-pitch aerodynamic coupling relationship that causes the model to crash. With the flight computer activated (bottom), there is no obvious roll-pitch relationship, and the aircraft does not crash. *Image created by Kevin Shen.*

- As the pilot, I observed that the model was difficult to control without the flight computer, and while it was still flyable, required skill and was **prone to coupled oscillations and crashes**, as seen in Figure 23.
- With the flight computer, the model flew much more **similarly to conventional model aircraft** and avoided coupling-induced crashes.



Figure 23. The oblique wing oscillating and crashing without flight computer assistance to correct the aerodynamic coupling motion. *Image created by Kevin Shen.*

- Statistically analyzing control authority (the ability of the pilot to independently control each axis) to provide an additional perspective on improved stability and control, supplementing previous coupling analyses.

Figure 24. Improved control authority shown by comparing R^2 values of aircraft response rate vs pilot command, with and without flight computer. R^2 was used to evaluate control authority because R^2 measures the strength of the relationship between pilot command and aircraft response. *Image created by Kevin Shen.*

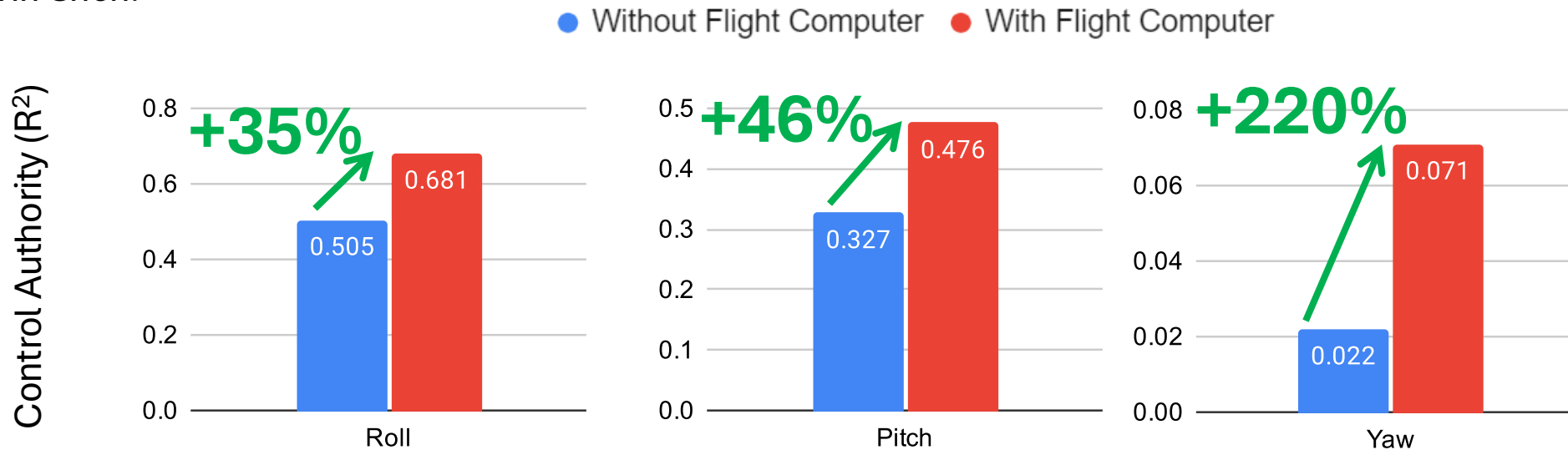
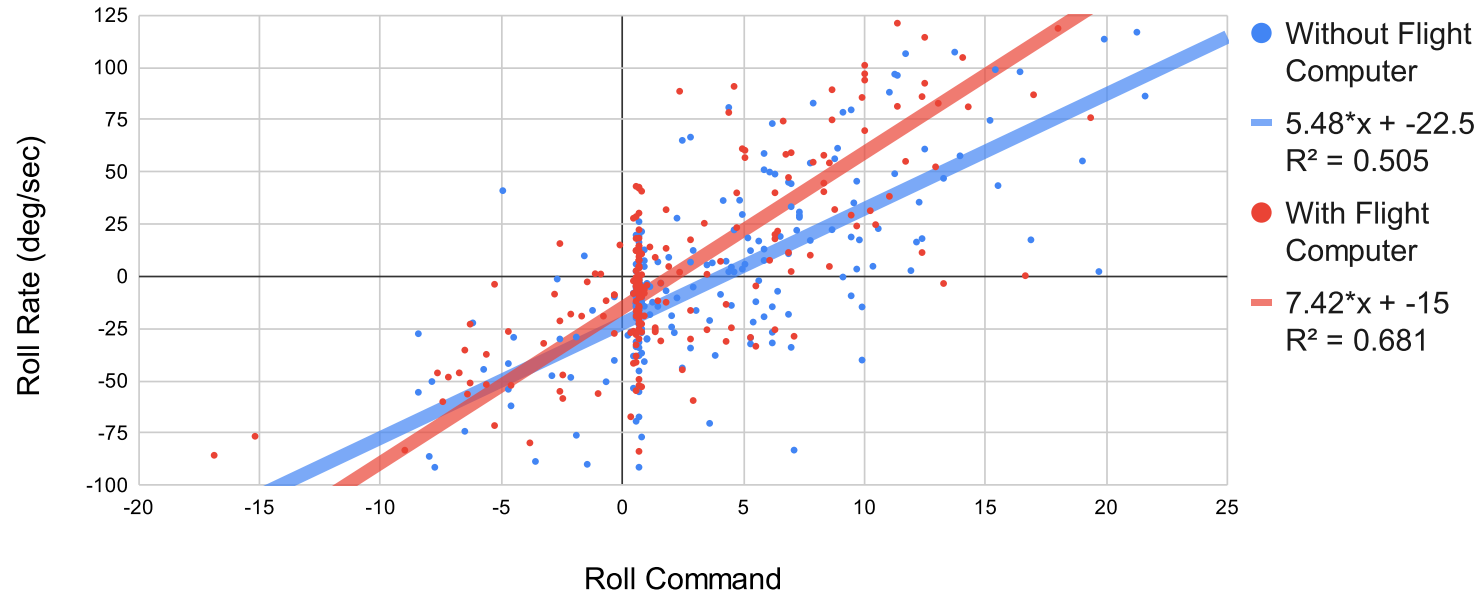


Figure 25. Scatterplot of roll rate versus roll command as an example of the source of data for Figure 19. Notice how the regression line for data from flights with the flight computer has not only a higher R^2 , but also has a higher slope, further indicating increased controllability. *Image created by Kevin Shen.*



- Data show that just by implementing the anti-coupling software, control authority was **improved on all 3 axes by at least 34%.**
- Because the only change was the implementation of the anti-coupling flight algorithm, the increase in control authority is a further indicator of reduced coupling, as each pilot command impacts more directly its intended axis without affecting others.

- **Limitation:** The anti-coupling flight computer is currently tuned for only the specific flight conditions of this model, limiting its adaptability.
 - ❑ **Future Work:** Study how airspeed, altitude, and other variables impact flight control, and implement gain scheduling if needed.

- **Limitation:** The model was entirely home-built and self-funded, limiting quality of materials, manufacturing techniques, and model size.
 - ❑ **Future Work:** Raise funding to build a larger and more precisely built oblique wing model for more accurate flight testing.

- ✈ The design, optimization, and analysis of an **aerodynamic model** suggested that oblique-wing aircraft could save up to **96 million tonnes** of CO₂ emissions and **25 billion dollars** every year.
- 🎯 The **new anti-coupling flight computer** significantly reduced aerodynamic coupling in roll-pitch by **99%**. Multiple statistical analyses demonstrated the oblique wing model was more **safe and controllable**.
- 🍃 This study urges further research into a modern implementation of the oblique wing in airliners and transports, contributing to more **sustainable aviation**.

*“That oblique-wing vehicles will fly is not in doubt. The only question is **when**.”*

— Bruce Larrimer, NASA Aeronautics. [4]

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All remaining references can be found in the full paper.